

WORKS WELL FOR
Continuous traffic

WATCH OUT FOR
Guard bands waste spectrum



WORKS WELL FOR
Scheduled voice & data

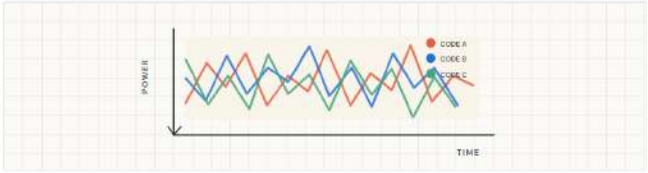
WATCH OUT FOR
Needs tight synchronization

03 CDMA

SPLIT BY CODE

CODE DIVISION MULTIPLE ACCESS

Everyone transmits at once; unique codes separate each user's signal.



WORKS WELL FOR
Many simultaneous users

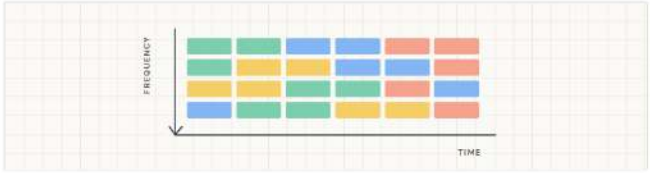
WATCH OUT FOR
Complex signal processing

04 OFDMA

SPLIT BY TIME + FREQUENCY

ORTHOGONAL FREQUENCY DIVISION MULTIPLE ACCESS

Small time-frequency blocks are assigned dynamically to each user.



WORKS WELL FOR
High-speed mobile data

WATCH OUT FOR
Sensitive to frequency offset

SAME GOAL — DIFFERENT WAY TO SHARE THE CHANNEL

USER 1 · USER 2 · USER 3 · USER 4

WIRELESS COMMUNICATION · VISUAL GUIDE 10

MULTIPLE ACCESS, FOUR WAYS.

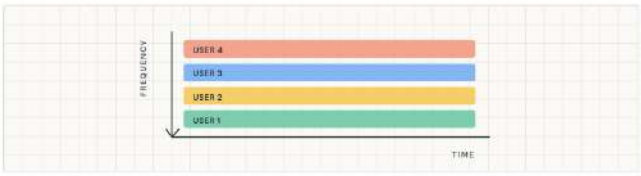
How many users share the same communication medium—separated by **frequency, time, code,** or a flexible mix of both.

01 FDMA

FREQUENCY DIVISION MULTIPLE ACCESS

SPLIT BY FREQUENCY

Each user gets a dedicated frequency band for the entire session.



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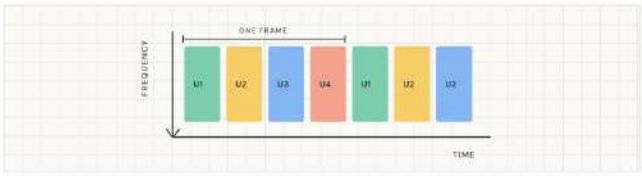
Guard bands waste spectrum

02 TDMA

TIME DIVISION MULTIPLE ACCESS

SPLIT BY TIME

Users share one channel, taking turns in repeating time slots.



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WATCH OUT FOR

Needs tight synchronization